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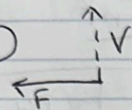
Midterm #2

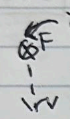
Unit 4

① direction of B-field a-c, w $\vec{v}$

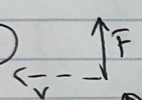
lorentz force : $\vec{F} = q(\vec{v} \times \vec{B})$

ⓐ 

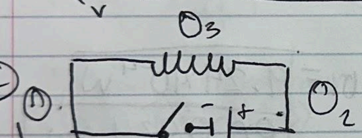
right hand rule: $\boxed{\text{B-field is into the page.}}$

ⓑ 

$\boxed{\text{B-field is left (negative)}}$

ⓒ 

$\boxed{\text{B-field is out of the page.}}$

② 

ⓐ direction of current in coil 1 is clockwise / B-field is created to the left.
- after a long time there'd be a constant flow, so there'd be no induced current
- when it's opened is opposite of when it's closed so that'd be counterclockwise.

ⓑ for coil 2:
- when close: counterclockwise, opposite side of coil 1.
- constant flow means no induced current
- when oppened: clockwise (opposite of closed.

ⓒ for coil 3: - no current induced for open, closed, or ~~open~~ closed for a long time because it's in the middle & above the Solenoid.

③ ⓐ loop → 2.0T field. emf on ring: d = 2.2cm, .25s into field.

$B = 2T,\ d = 0.022\,m,\ \Delta t = 0.25s$

$\mathcal{E}_{av} = \frac{\Delta \Phi_B}{\Delta t} \quad \Phi_B = B \cdot A \cdot \cos\theta$

$\mathcal{E}_{av} = \frac{2 \times (\pi)(0.011)^2\ T \cdot m^2}{0.25s} = \frac{2 \times 3.01 \times 10^{-4}\ Tm^2}{0.25s}$

$= \frac{7.602 \times 10^{-4}\ Wb}{0.25s} = \boxed{3.04 \times 10^{-3} V}$ or $\boxed{3.04 mV}$

ⓑ current if resistance of ring is 0.01 Ω?

$I = \frac{\mathcal{E}}{R} = \frac{3.04 \times 10^{-3} V}{0.01\Omega} = \boxed{0.304 A}$

ⓒ av. power dissipated into ring?

$P = I^2 R$

$P = (0.304)^2 (0.01) = 0.0924 \cdot 0.01 = \boxed{9.24 \times 10^{-4} W}$ $\boxed{\text{or } 0.924 mW}$

④ ⓐ 1875 rad/s w/ 18V emf peak. 1.0 × 3.0 cm coil in a 0.64 T field. How many turns?

$N = \frac{\mathcal{E}_{peak}}{BA\omega} = \frac{18V}{0.64 \times (0.01 \times 0.03) \times 1875} = \frac{18}{\cancel{675}\ 0.36} = \boxed{\cancel{\ldots}}$ $\boxed{50 \text{ turns}}$

ⓑ what's the period of AC output of the generator?

$f = \frac{\omega}{2\pi} = \frac{1875}{2\pi} = 298.4 Hz$

$T = \frac{1}{f} = \frac{1}{298.4} = \boxed{3.35 ms \text{ or } 3.35 \times 10^{-3} s}$

⑤ ⓐ 240 V to 120 V - What's the ratio of turns?

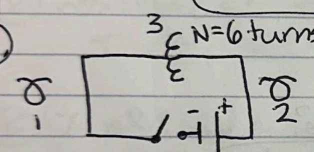
$\frac{V_p}{V_s} = \frac{N_p}{N_s}$ s = secondary, p = primary

$\frac{240}{120} = \frac{N_p}{N_s} = 2$, so 2x more turns in the primary coil   2:1 ratio

ⓑ input to output current?

$\frac{I_p}{I_o} = \frac{1}{2}$ input current is $\frac{1}{2}$ of output current, so it's a 1:2 ratio

⑥



ⓐ $V(t) = 120\sin(240\pi t)$

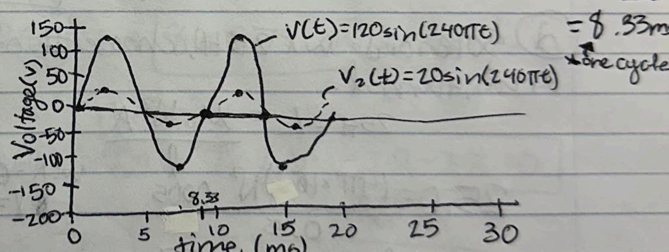
amp = ±120V $\omega = 240\pi$

$f = \frac{\omega}{2\pi} = \frac{240\pi}{2\pi} = 120$ Hz $T = \frac{1}{f} = \frac{1}{120} = 0.0083$ sec. = 8.33 ms

★ one cycle

★ now a transformer

$V(t) = V_0 \sin(\omega t)$



ⓑ add induced emf of coil 3

$\frac{\text{coil 3}}{\text{solenoid}} = \frac{1}{6}$

$V_2(t) = \frac{1}{6} V_0 \sin(\omega t)$

$= 20\sin(240\pi t)$ V

amp = ± 20 $\omega = 240\pi$

T = 8.33 ms

ⓒ if $V_0 = 120$V & $\omega = (240\pi)$ Hz, when does $V(t) = 0$?

$V(t) = V_0 \sin(\omega t)$

$V(t) = 120\sin(240\pi t)$

$0 = 120\sin(240\pi t)$

$240\pi t = n\pi$, $t = \frac{n}{240}$

$\boxed{t = \frac{1}{240} = 0.004175 = 4.17\text{ms}}$ → every 4.17ms, t = 0 again.

★ starts w/ 0, 4.17, ...

(7) $t=?$, $A=0.1$, 2.0 mH inductor, 500V emf

Faraday's Law: $\mathcal{E}=L\cdot\frac{\Delta I}{\Delta t}$ $\quad \mathcal{E}=500V$, $L=2mH$, $\Delta I=0.1A$

$$\Delta t=\frac{L\cdot\Delta I}{\mathcal{E}}$$

$$\Delta t=\frac{(2\times10^{-3})\times0.1}{500}$$

$$=\frac{2\times10^{-4}}{500}=\boxed{4.00\times10^{-7}s=400ns}$$

(8) (a) 25H, emf = ?, 100A off in 80 ms.

$$\mathcal{E}=L\cdot\frac{\Delta I}{\Delta t}=25\times\frac{100}{0.08}=\boxed{31{,}250V=31.25kV}$$

(b) energy at full current?

$$E=\frac{1}{2}LI^2$$

$$E=\frac{1}{2}(25)(100)^2=\boxed{125{,}000J}$$

(c) what rate ~~in~~ Watts dissipated for 80ms?

$$P=\frac{E}{\Delta t} \quad P=\frac{125{,}000J}{0.08s}=\boxed{1.56\times10^6\ Watts}$$

(d) Solenoid w/ 25H, choose # of turns N & A (cross-sectional area), turns l.

*so #s just have to make L=25!

$$25H=L=\frac{\mu_0N^2A}{l}$$

$$25.0=\frac{(4\pi\times10^{-7})N^2\ 0.005}{0.5} \quad w: A=0.005m^2,\ l=0.5m \quad N=?$$

$$N^2=\frac{Ll}{\mu_0A} \quad N=\sqrt{\frac{Ll}{\mu_0A}} \rightarrow N=\sqrt{\frac{25(0.5)}{4\pi\times10^{-7}(0.005)}}$$

$$N=\sqrt{\frac{12.5}{6.283\times10^{-9}}}=\sqrt{1.99\times10^9}=N=44{,}600\text{ turns}$$

$$\boxed{\text{A valid solenoid with } L=25.0H \text{ could have } A=0.005m^2,\ l=0.5m \text{ and } N=44{,}600 \text{ turns.}}$$

ⓒ What is the reactance of the inductor at 10 kHz? $\rightarrow \Omega$

$X_L = 2\pi f L \qquad L = 25.0\,H$

$X_L = 2\pi(10{,}000)(25)$

$X_L = 1{,}570{,}000 = \boxed{1.57\times10^6\,\Omega}$

⑨ ⓐ RL circuit, 20ns, resistance of 5MΩ, inductance?

$\tau = \frac{L}{R} = \quad L = \tau R$

$L = (20\times10^{-9}s)(5\times10^6) = 100\times10^{-3} = \boxed{0.1\,H}$

ⓑ resistance = ?, time: 1.0 ns

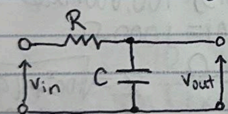
$\tau = \frac{L}{R} \quad R = \frac{L}{\tau} = \frac{0.1}{1\times10^{-9}} = \boxed{1.0\times10^8\,\Omega \text{ or } 100M\Omega}$

ⓒ % of final current I_0 to flow after deactivation, 30ns

$I(t) = I_0 e^{-t/\tau} \qquad \frac{I(t)}{I_0} = e^{-t/\tau}$

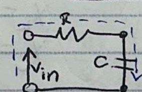
$e^{-3.0/20} = e^{-0.15} = 0.8607 = \boxed{86.07\%\text{ of } I_0 \text{ left}}$

⑩



ground at 0V

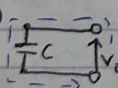
ⓐ



$\boxed{V_{in} - IR - IZ_c = 0}$

$Z_c = \frac{1}{j\omega C}$

ⓑ



$\boxed{V_{out} = IZ_c}$

ⓒ $\boxed{\begin{array}{l} V_{in} = I\left(R + \frac{1}{j\omega C}\right) \\ V_{out} = I\left(\frac{1}{j\omega C}\right)\end{array}}$

ⓓ $f = 100(2\pi RC)^{-1}: \ 100 \cdot \frac{1}{2\pi RC} = \frac{100}{2\pi}$ $\quad \omega = 100\cdot\omega_0$

$\frac{V_{out}}{V_{in}} = \frac{\cancel{I}\left(\frac{1}{j\omega C}\right)}{\cancel{I}\left(R + \frac{1}{j\omega C}\right)} = \boxed{\frac{1}{1+j\omega RC}}$

$\frac{V_{out}}{V_{in}} = \frac{1}{1+j(100)}$

$= \frac{1}{\sqrt{1+100^2}} = \frac{1}{\sqrt{10001}} = 0.00999$ ~1% of input signal @ ↑ freq

$f = 0.1(2\pi RC)^{-1}: \ 0.1 \frac{1}{2\pi RC} \quad \omega = 0.1\omega_0$

$\frac{V_{out}}{V_{in}} = \frac{1}{1+j0.1} = \frac{1}{\sqrt{1+0.1^2}} = \frac{1}{\sqrt{1.01}} = 0.995$ ~99% input @ low freq.

11 ⓐ $f_0 = ?$, $R = 0.1\,k\Omega$, $C = 1\mu F$, $L = 10mH$.

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10\times10^{-3}\cdot1\times10^{-6}}} = \frac{1}{2\pi\sqrt{10^{-8}}} = \frac{1}{2\pi\times10^{-4}}$$

$$f_0 = \frac{1}{6.283\times10^{-4}} = \boxed{1591.5\,Hz}$$

ⓑ impedance (Z) at $f_1 = 0.1 f_0$ & $f_2 = 10 f_0$ $\quad \omega = 2\pi f$

$f_1 = 0.1 f_0 = 159.15\,Hz$

$Z = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$

$\omega = 2\pi(159.15)$
$\omega = 1000\,rad/s$
$X_L = \omega L = 10\,\Omega$
$X_C = \frac{1}{\omega C} = 1000\,\Omega$
$X = X_L - X_C = -990\,\Omega$

$Z = \sqrt{100^2 + (-990)^2}$

$Z = \sqrt{990100}$

$\boxed{Z = 995\,\Omega}$

$f_2 = 10 f_0 = 15915\,Hz$

$\omega = 2\pi(15915)$
$\omega = 100,000\,rad/s$
$X_L = \omega L = 1000\,\Omega$
$X_C = \frac{1}{\omega C} = 10\,\Omega$
$X = X_L - X_C = 990\,\Omega$

$Z = \sqrt{100^2 + 990^2}$

$Z = \sqrt{990100}$

$\boxed{Z = 995\,\Omega}$

12 ⓐ $V_{rms} = 120V$, $I_{rms} = ?$, $10f_0$ & $0.1 f_0$ $\quad R = 100\,\Omega \quad C = 1\times10^{-6}F$
$L = 0.01H \quad f_0 = \frac{1}{2\pi\sqrt{LC}} = 1591.5\,Hz$

$I_{rms} = \frac{V_{rms}}{Z}$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$X_L = \omega L$, $X_C = \frac{1}{\omega C}$

for $10f_0$: $15,915\,Hz$

$\omega = 2\pi f = 2\pi(15,915) = 100,000\,rad/s$

$X_L = \omega L = 100000 \cdot 0.01 = 1000\,\Omega$

$X_C = \frac{1}{100000(1\times10^{-6})} = 10\,\Omega$

$Z = \sqrt{100^2 + (990)^2}$

$Z = 995\,\Omega$

$\boxed{I_{rms} = \frac{120V}{995\Omega} = 0.121A}$

for $0.1 f_0$: $159.15\,Hz$

$\omega = 2\pi(159.15) = 1000\,rad/s$

$X_L = 1000(0.01) = 10\,\Omega$

$X_C = \frac{1}{1000\times10^{-6}} = 1000\,\Omega$

$Z = \sqrt{100^2 + (-990)^2}$

$Z = 995\,\Omega$

$\boxed{I_{rms} = \frac{120V}{995\Omega} = 0.121A}$

ⓑ $P_{rms} = ?$

$P_{rms} = V_{rms} I_{rms} \cos\phi$

$\tan\phi = \frac{X_L - X_C}{R}$

$\cos\phi = \frac{R}{Z}$

$\cos\phi = \frac{100}{995} = 0.1005$

$P_{rms} = 120 \cdot 0.121 \cdot 0.1005$

$\boxed{P_{rms} = 1.46W \text{ for } 0.1f_0 \text{ \& } 10f_0}$

(13) resonance freq: 1.4 MHz, audio: 10 kHz, what freq (3)
- 3 frequencies are: $\boxed{\text{carrier, upper sideband, \& lower side band}}$

carrier/resonance: f_c
$f_c = 1.4\,MHz$
$\boxed{f_c = 1{,}400{,}000\,Hz}$

upper sideband: $f_c + f_m$
$f_{us} = f_c + f_m$
$= 1.4\times10^6 + 1.0\times10^4\ Hz$
$\boxed{f_{us} = 1{,}410{,}000\,Hz \text{ or } 1.41\times10^6\,Hz}$

lower sideband: $f_c - f_m$
$f_{ls} = f_c - f_m$
$= 1.4\times10^6 - 1.0\times10^4$
$\boxed{f_{ls} = 1{,}390{,}000\,Hz \text{ or } 1.39\times10^6\,Hz}$

Unit 5

(1) (a) B-field 1cm laterally, charges 0-200 nC in 2 μs

$B = \frac{\mu_0 I_d}{2\pi r}$ $I_d = \frac{\Delta Q}{\Delta t}$ $\Delta Q = 200\,nC = 2\times10^{-7}\,C$ $r = 1\,cm = 1\times10^{-2}\,m$
$\Delta t = 2\,\mu s = 2\times10^{-6}\,s$

$$B = \frac{(4\pi\times10^{-7})\left(\frac{2\times10^{-7}}{2\times10^{-6}}\right)}{2\pi(1\times10^{-2})} = \frac{4\times10^{-7}(0.1)}{2\times10^{-2}} = \boxed{2.0\times10^{-6}\,T = 2.0\,\mu T}$$

(b) What "current" is responsible for B-field?
- instead of conduction current it's displacement current. Displacement current still generates a magnetic field. It is: $I_d = \frac{\Delta Q}{\Delta t} = \frac{2\times10^{-7}}{2\times10^{-6}} = \boxed{0.1\,A}$

2 a $10\mu s$, what's displacement? (pulse = there & back)

$D = c \cdot \Delta t$ c = speed of light $= 3\times10^8$ m/s

$\Delta t = 10\mu s = 10\times10^{-6} s$

$D = 3\times10^8(10\times10^{-6})$

$D = 3000$ m round trip

$\boxed{d = 1500\text{ m from aircraft to ice}}$

b What fraction of power, standard index of refraction @RF

Fresnel reflection: $R = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2$

Standard RF refraction:

- Air $= 1.00$ (n_1)
- ice $= 1.78$ (n_2)

$R = \left(\frac{1 - 1.78}{1 + 1.78}\right)^2 = \left(\frac{-.78}{2.78}\right)^2 = (-0.2806)^2 = \boxed{0.0787}$

$\boxed{\text{So, 7.87\% of power is reflected}}$

3 a 1kW onto $10 cm^2$ at distance of 1m. Intensity?

$I = \frac{P}{A} = \frac{1000 W}{10\times10^{-4} m^2} = \boxed{1.0\times10^6 W/m^2}$

$10 cm^2 = 10\times10^{-4} m^2$

b if intensity ↓ as distance squared, what's I at 2m

$I_2 = I_1\left(\frac{r_1}{r_2}\right)^2$ $r_1 = 1m$, $r_2 = 2m$

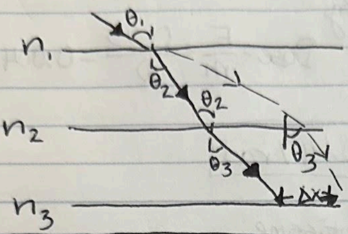
$I_1 = 1\times10^6 W/m^2$

$I_2 = 1.0\times10^6 \cdot \left(\frac{1}{2}\right)^2$

$I_2 = 1\times10^6 \left(\frac{1}{4}\right)$

$\boxed{I_2 = 2.5\times10^5 W/m^2}$

(4) (a) prove that $\theta_3 = \theta_1$ in figure 4. if $n_1 = n_3$.



$\theta_3 = \theta_1$ if $n_1 = n_3$

$n_1 \sin\theta_1 = n_2 \sin\theta_2$

$n_2 \sin\theta_2 = n_3 \sin\theta_3$

$n_1 \sin\theta_1 = n_2 \sin\theta_2 = n_3 \sin\theta_3$

$\boxed{\text{so, } n_1\sin\theta_1 = n_3 \sin\theta_3\text{. if } n_1 = n_3 \text{, then } \sin\theta_1 = \sin\theta_3\text{. so, since } n_1 = n_3,\ \theta_1 = \theta_3.}$

(b) 12cm lens 8cm from ear, magnification=?

$M = \frac{-d_i}{d_o}$ $\qquad \frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \rightarrow \frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$

$\frac{1}{f} = \frac{1}{12}$ $\qquad = \left(\frac{1}{12}\right)^{\times 2} - \left(\frac{1}{8}\right)^{\times 3} = \frac{2}{24} - \frac{3}{24} = \frac{-1}{24} = \boxed{-24\text{cm} = d_i}$

$d = 8,\ \frac{1}{d_o} = \frac{1}{8}$

$M = \frac{-(-24\text{cm})}{8} = \boxed{3.00\times \text{ magnification}}$

(5) 200 barns at 10 KeV, 1cm thick, what fraction?

$I = I_0 e^{-n\sigma x}$ $\qquad$ n=density, σ=cross section, x=thickness

$\sigma = 200 \text{ barns} = 200\times10^{-28}\,m^2 = 2.0\times10^{-26}\,m^2$

$x = 0.01m$

lead density $\rightarrow n = \frac{\rho N_A}{A} = \frac{11,340\ kg/m^3\ (6.022\times10^{23}\ \text{atom/mol})}{0.2072\ kg/mol} = 3.29\times10^{28}\ \text{atoms}/m^3$

$\frac{I}{I_0} = e^{-3.29\times10^{28}\cdot 2\times10^{-26}\cdot 0.01} = e^{-6.56} = 0.00139 = \boxed{1.39\times10^{-3}}$

so, 0.139% passes through the rest.

⑥ ⓐ 250mJ into 60kg, radioactive dose?

$1\,rad = \frac{0.01J}{1kg} = 0.01 J/kg$

$250mJ = 0.25 J$

$Dose = \frac{E}{m} = \frac{0.25}{60} = 0.00417 J/kg$

$\text{dose in rads} = \frac{0.00417}{0.01} = \boxed{0.417\,rad}$

ⓑ concentration in 2kg of tissue

$250mJ = 0.25J$ $\quad dose = \frac{0.25}{2} = \frac{0.125}{0.01} = \boxed{12.5\,rads}$

ⓒ RBE=1, dose?

Dose in Sv = Dose in Gy × RBE

* 1 Gy = 100 rads

0.125 Gy = 12.5 rads

RBE = 1

$\text{Dose in } Sv = 0.125 \times 1 = \boxed{0.125\,Sv}$

ⓓ a serious health risk?

according to Google: 0.1–0.5Sv = mild, temp. sympt., >1Sv = radiation sickness, >4–5Sv = potentially lethal

-so, since the dose is 0.125Sv, there's likely symptoms and can impact long term cancer risks, BUT it isn't life threatening. This is above the saftey limit for a 1 time exposure.

Thank you for a wonderful year of physics!
-Emily